

IN-SPACe Model Rocketry India Student Competition 2026

**Funding & Joint Proposal by
Students from**

**Department of Aerospace Engineering,
Department of CSE,
Department of EECE ,
Department of Mechanical engineering,
GSCE,GSCSE
GITAM DEEMED TO BE UNIVERSITY,
GITAM Hyderabad.**

Abstract

This proposal seeks institutional permission, funding and sponsorship support from GITAM Deemed to be University under the Student Innovation Plan (SIP) for the participation of the GITAM inaugural student team in the IN-SPACE Model Rocketry India Student Competition 2026 (2nd Edition), organized by the Indian National Space Promotion and Authorization Centre (IN-SPACe), Government of India.

This is a historic first in GITAM's academic journey — the Department of Aerospace Engineering is fielding its first competitive rocketry team at a nationally recognized, government-organized space technology event. The competition, conducted under the auspices of the Department of Space, Government of India, challenges undergraduate teams to design, fabricate, integrate, and launch a custom solid-propellant rocket capable of delivering a 1.0 kg satellite-class payload (CAN-7U) to a target altitude of 1,000 metres, with full telemetry, autonomous recovery, and post-flight analysis.

Participation in this competition directly aligns with GITAM's strategic vision of excellence in technology, innovation, and national contribution. It will provide students hands-on experience in aerospace systems engineering, propulsion, avionics, structures, and mission operations — skills that are critical to India's rapidly growing space sector.

The team respectfully requests a total funding support of ₹1,50,000 (Rupees One Lakh fifty thousand Only) from the institution, along with institutional facilitation in connecting with potential corporate sponsors. The funds will cover mandatory competition fees, workshop costs, component procurement, avionics development, structural fabrication, launch station and ground station infrastructure.

The Ask: Institutional Support Required

KEY HIGHLIGHTS:

- GITAM's FIRST-EVER participation in a national government-organized competition
- Organized by IN-SPACe, Department of Space, Govt. of India
- 70 teams from 18 states participated in 1st Edition (2024-25)

Funding Source	Amount	
GITAM Student Innovation Plan (SIP)	₹1,50,000	
Corporate Sponsorships (through GITAM)	₹1,66,802	
Grand Total Project Cost	₹3,16,802	

INTERDISCIPLINARY DREAM TEAM

Engineering Branch	Responsibilities
Aerospace Engineering	Rocket design, aerodynamics, mission planning
ECE (Electronics)	Avionics, telemetry, PCB design
Mechanical Engineering	Structural analysis, fabrication, recovery systems
CSE (Computer Science)	Ground station software, data logging, simulation

The Technical Challenge – Mission Parameters

Parameter	Specification
Target Altitude	1,000 m $\pm$ 100 m
Payload Mass	1.0 kg $\pm$ 0.05 kg (CAN-7U format)
Max Rocket Length	< 180 cm (incl. fins)
Motor Class	K-class ($\leq$ 2800 Ns) from IN-SPACE approved vendors
Recovery	Autonomous parachute, descent rate 2-5 m/s
Telemetry	Mandatory onboard logging + real-time downlink

HISTORIC FIRST for GITAM

This is a watershed moment in GITAM's academic journey:

- FIRST TIME any GITAM team is participating in a government-organized
- national rocketry competition
- FIRST TIME Department of Aerospace Engineering is fielding
- a competitive rocketry team
- We are PAVING THE PATH for future generations of GITAM aerospace students
- **WHAT THIS MEANS:**
 - GITAM's name alongside IITs, NITs at a national platform
 - Creating institutional memory and knowledge transfer
 - Building a legacy for future teams
 - Demonstrating GITAM's capability in India's space sector

Competition Roadmap – April to Nov 2026

Phase	Activity	Deadline
Phase 1	Registration & Institutional Approval	30 Apr 2026
Phase 2	PDR Submission & Presentation	1 May – 15 Jul 2026
Phase 3	IN-SPACE Rocket Modelling Workshop	1–16 Aug 2026
Phase 4	CDR Submission & Review	17 Aug – 10 Oct 2026
Phase 5	Flight Readiness Review (FRR)	Oct-Nov 2026
Phase 6	Grand Finale Launch	Oct-Nov 2026

Why This Matters – Institutional Alignment

- **NATIONAL EDUCATION POLICY (NEP) 2020**
 - Experiential, project-based learning in aerospace engineering
- **NATIONAL SPACE POLICY 2023**
 - Direct response to Govt. of India's workforce development
- **NATIONAL VISIBILITY**
 - GITAM alongside IITs, NITs at government-organized platform
- **STUDENT SKILL DEVELOPMENT**
 - Propulsion, avionics, telemetry, mission operations
 - Directly applicable to ISRO, DRDO, and private space sector
- **RESEARCH OUTPUT**
 - Technical reports, design data, potential peer-reviewed publications

Budget Summary – Total: ₹3,16,802

Category	Subtotal (₹)
1. Mandatory Fees + Propulsion	1,21,905
2. Structural & Mechanical Components	33,879
3. Payload System – CAN-7U Satellite	26,609
4. Avionics & Electrical Systems	27,827
5. Recovery System	32,804
6. Ground Support Equipment	20,408
7. Fabrication & Labor	17,700
8. Testing & Contingency	35,670
GRAND TOTAL	3,16,802

Funding Sources – Multiple Revenue Streams

Source	Expected Amount	Status
GITAM Student Innovation Plan (SIP)	₹2,00,000	Requested
GITAM Venture Development Centre (VDC)	₹1,00,000 - ₹1,50,000	To be pitched
Corporate Sponsorships	₹1,16,802	Via GITAM facilitation
TOTAL EXPECTED	₹4,16,802 - ₹4,66,802	

Budget Detail 1: Mandatory Fees + Propulsion (₹1,21,905)

Item	Cost (₹)
Competition Registration Fee (Phase 1)	10,000
Rocket Modelling Workshop – Participant 1	25,000
Rocket Modelling Workshop – Participant 2	25,000
CDR Submission Fee (Phase 4)	15,000
K-Class Motor Reload (incl. HAZMAT)	22,420
54mm Motor Casing Hardware	24,485

Budget Detail 2: Structural & Mechanical (₹33,879)

Component	Cost (₹)
Airframe Body Tube (Al 6061-T6)	2,832
Fin Sheet Stock (Al 6061-T6)	1,699
Nose Cone (PETG/Nylon 3D print)	14,160
Motor Mount + Centering Rings	2,950
Payload Bay + Bulkheads	3,540
Kevlar Shock Cord + Tether	2,950
Other hardware + adhesives	5,748

Budget Detail 3: Payload System – CAN-7U (₹26,609)

Component	Cost (₹)
CAN-7U Chassis	5,900
Stratologger CF Altimeter (MANDATORY)	9,086
Payload Telemetry – XBee Pro S2C	3,894
Payload Parachute (24 inch)	2,773
Li-ion 18650 Battery Pack	1,770
Environmental Sensor Module	1,770

Budget Detail 4: Avionics & Electrical (₹27,827)

Component	Cost (₹)
Flight Computer (Teensy 4.1 / STM32)	3,540
NavIC GNSS Receiver (L5/L1 dual-band)	5,310
XBee Pro S2C Telemetry	3,894
Barometric Altimeter (Primary + Backup)	746
3-Axis Accelerometer	1,416
Power Distribution + Custom PCB	4,130
Wiring, connectors, batteries	8,791

Budget Detail 5: Recovery System (₹32,804)

Component	Cost (₹)
Main Parachute – 60 inch	20,355
Payload Parachute – 24 inch	2,773
CO2/Spring Deployment Mechanism	2,360
Kevlar Shock Cord + Tether Lines	3,776
HAM Beacon / APRS Tracker	3,540

Budget Detail 6: GSE + Fabrication + Contingency (₹73,778)

Category	Item	Cost (₹)
Ground Support	XBee Ground Station + Antenna	5,664
Ground Support	Launch Rail (6m 4040 T-Slot)	4,602
Ground Support	Blast Deflector + Support	3,894
Ground Support	Fire Extinguisher + Power	3,776
Fabrication	3D Printing (Nose cone + parts)	14,160
Fabrication	Electronics Assembly Labor	3,540
Testing	Scaled Prototype Rocket	3,540
Testing	Vibration/Shock Tests	2,360
Contingency	Reserve (~10% of hardware)	28,000

IN-SPACe Mandatory Requirements – Compliance

Requirement	Status
Altitude: 1,000 m $\pm$ 100 m	
Payload: 1.0 kg $\pm$ 0.05 kg CAN-7U	
Single stage only	
Length < 180 cm	
Motor from approved vendor	
Stratologger CF altimeter	
No LiPo batteries – 18650 only	
No spring contacts	
No outsourcing (student-designed)	
Team size:	

Design:

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Rocket design Motors & Configuration Flight simulations

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- Sustainer
 - Nose Cone
 - Body Tube
 - Shock Cord 1
 - Shock Cord 2
 - Parachute
 - Transition
 - Mass Component
 - Parachute
 - Shock Cord
 - Body Tube
 - Freeform Fin Set 1deg
 - Trapezoidal Fin Set 2deg

Move up

Move down

Edit

Duplicate

Delete

Add new component

Freeform

Tube Fins

Rail Button

Launch Lug

Inner Components

Inner Tube

Tube Coupler

Centering Ring

Bulkhead

Engine Block

Mass Components

Parachute

Streamer

Shock Cord

Mass Component

View Type: 3D Figure

Zoom: Fit (6.4%)

Stability: ☒ Show CG/CP

Stages: Sustainer

Flight configuration: [K650T-P]

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Length 173 cm, max. diameter 21.6 cm

Mass with no motors 8075 g

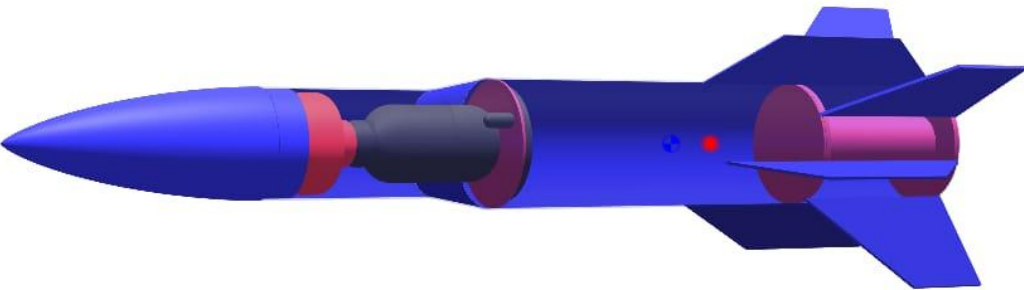
Mass with motors 11010 g

Stability: 0.372 cal / 4.66 %

CG: 116 cm

CP: 124 cm

at M=0.300



Flight configuration: [K650T-P]

Apogee: 1022 m

Max. velocity: 161 m/s (Mach 0.476)

Max. acceleration: 56.1 m/s²

Click to select Shift+click to select other Double-click to edit Click+drag to move

☒ Show warnings

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- Sustainer
 - Nose Cone
 - Body Tube
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 - Freeform Fin Set 1deg
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Add new component

Move up Move down

Edit Duplicate Delete

Inner Components

- Freeform
- Tube Fins
- Rail Button
- Launch Lug
- Inner Tube
- Tube Coupler
- Centering Ring
- Bulkhead
- Engine Block

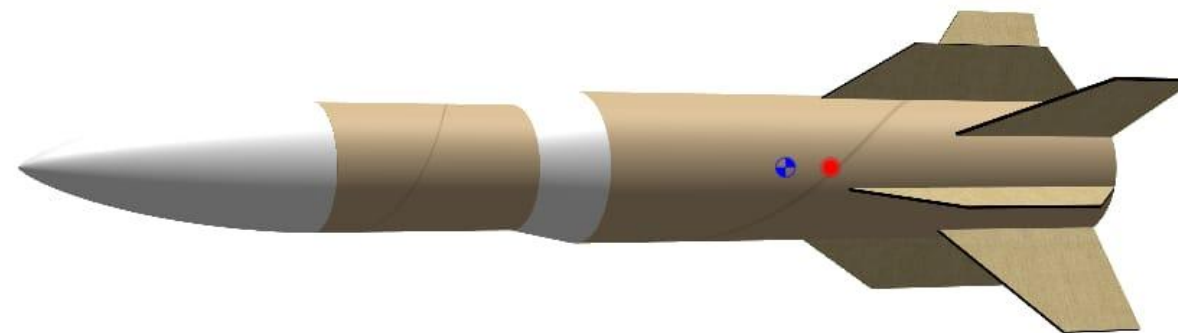
Mass Components

- Parachute
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View Type: 3D Finished Zoom: Fit (6.4%) Stability: ☒ Show CG/CP Stages: Sustainer Flight configuration: [K650T-P]

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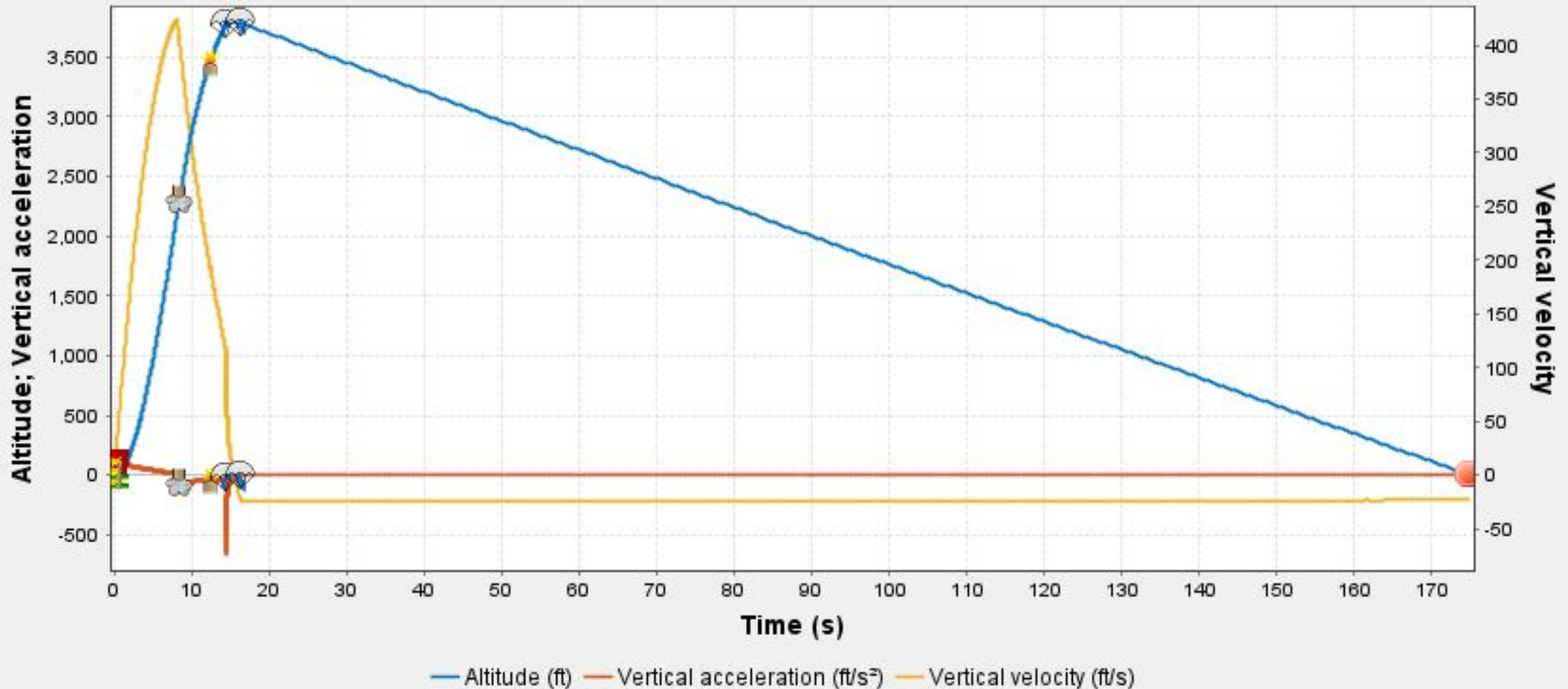


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Simulation Analysis:

Simulation 1

Vertical motion vs. time



What We Need from GITAM

- **1. FINANCIAL SUPPORT (SIP Grant): ₹**
 - Mandatory fees + propulsion hardware + critical components
- **2. VDC PITCH SUPPORT**
 - Endorsement for Venture Development Centre pitch
 - Expected funding:
- **3. SPONSORSHIP FACILITATION (60% of remaining budget)**
 - Official GITAM endorsement letter
 - Introductions to Industry Partnership network
- **4. DEPARTMENTAL SUPPORT**
 - Lab access for testing and integration
 - Faculty mentorship (up to 2 external mentors)
 - Computing resources (FEA, flight simulation)
 - Official approval letter (deadline: 30 April 2026)
 - Dedicated workspace for rocket assembly

Declaration & Request for Approval

- **WE DECLARE:**
 - All design, development by undergraduate students
 - No outsourcing (disqualification offence per IN-SPACE)
 - Full financial accountability
 - We will document everything to PAVE THE PATH for future teams
- **SEEKING APPROVAL FOR:**
 - ₹ under Student Innovation Plan (SIP)
 - VDC pitch endorsement
 - Sponsorship facilitation
 - Departmental lab access and workspace
- **NEXT STEPS:**
 - Sign-off from HoD, Aerospace Engineering
 - Approval from Dean, Research & Innovation
 - VDC pitch preparation
 - Official GITAM endorsement letter (deadline: 30 April 2026)

Thank you.

***Students of Department of Aerospace Engineering, Department of CSE,
Department of EECE,
Department of Mechanical engineering, GSCE, GSCSE / GITAM Deemed to be University***